

SANJAY V

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PROFESSIONAL SUMMARY

Software Engineer with nearly 2 years of experience and a strong passion for Machine Learning and Data Science, demonstrated through hands-on projects in RAG systems, ML pipelines, predictive analytics, NLP and deep learning. Pursuing Data Scientist or ML Engineer opportunities.

TECHNICAL SKILLS

- **Programming Languages:** Python, R and Java Programming.
- **Machine Learning & Deep Learning:** TensorFlow, Keras, PyTorch, Scikit-Learn, Neural Networks, Time Series Forecasting.
- **Data Analytics:** Pandas, NumPy, Matplotlib, Seaborn, Statistical Analysis, A/B Testing, Hypothesis Testing.
- **Natural Language Processing:** NLTK, Transformers, Word2Vec, TF-IDF, Sentiment Analysis.
- **LLM & AI Applications:** HuggingFace, LangChain, LangGraph, MCP, RAG Architecture, LangSmith, FastAPI, Streamlit, Vector DB.
- **Business Intelligence:** Power BI, Tableau, Microsoft Excel, Alteryx, Data Visualization.
- **Databases & Cloud:** MySQL, MongoDB, BigQuery, Snowflake, Microsoft Azure, GCP.
- **Development Tools:** Git, Docker, Kubernetes, Spring Boot, Microservices, REST APIs, Agile/Scrum.

WORK EXPERIENCE

Software Engineer (Data Science / Analytics)

- Tech Mahindra, Bengaluru, India March 2024 - Present
- Developed multi-model time-series forecasting system (**ARIMA, SARIMA, Prophet, Exponential Smoothing**) with statistical validation (**ADF, T-Test, Chi-Square, ANOVA**) achieving **33%** improvement in accuracy (**MAPE/RMSE**); created automated **Power BI** dashboards enabling measurable cost reduction.
 - Performed statistical analysis (**ADF, T-test, Chi-Square** tests) on **500K+** records using **R Language** and built interactive dashboards with **ggplot2/dplyr** visualizations, enabling business analysts to identify real-time trends vs. static monthly reports.
 - Engineered ML classification model (**SVM, K-Neighbors** with **K=5**) using stratified **K-fold cross-validation** and **ROC-AUC** evaluation, achieving **91.4%** accuracy on hold-out test set; identified high-value sales leads with **92%** precision and **88%** recall.
 - Implemented customer segmentation using **K-Means** (elbow method, silhouette analysis) and **DBSCAN** clustering on RFM features; identified **3** clusters with top **40%** segment generating **60%** revenue; validated with **A/B testing** achieving **18%** higher engagement (**p < 0.05**).
 - Created Machine Learning NLP pre-processing pipeline (**NLTK Stemming, Lemmatization, TF-IDF** vectorization, **Word2Vec** embeddings) reducing dimensionality by **50%** and training time from **6** to **3** hours.
 - Built **Power BI (DAX), Tableau (LOD expressions)** dashboards and **Python** visuals (**Matplotlib, Seaborn**) reducing manual reporting by **80 hours/month**; freed analytics team to focus on **4** strategic initiatives.
 - Constructed end-to-end ETL pipeline using **Python (Pandas** validation, **NumPy** vectorization) and **SQL** (CTEs, window functions) on **100K+** records with anomaly detection (isolation forest, z-score methods), reducing data quality issues by **40%** and improving the training time.

KEY PROJECTS

Chat with Document: AI Assistant Document Reader

- LLM Application October 2025 - December 2025
- Developed an interactive **Streamlit-based AI Document Chatbot** that performs intelligent query routing via **LangGraph**—detecting intent and switching between **RAG-powered** answers or **base LLM reasoning**—reducing irrelevant responses by **40%**.
 - Engineered an end-to-end Retrieval-Augmented Generation (**RAG**) pipeline using **LangChain, HuggingFace** embeddings, and **Chroma** vector stores to extract knowledge from **100+** page PDFs, enabling **85%** accuracy in context-aware answers.
 - Orchestrated a production-ready **FastAPI** backend with **5+** modular **RAG** components and integrated **LangSmith** for evaluation, tracing, and observability across API calls, ensuring reliable and maintainable deployment.
- Additional Works: Expense Tracker** - Customized a **FastMCP** expense tracking API with **SQLite** backend using **Claude** for rapid development, handling **10+** categories with **CRUD** operations, summaries, and **MCP** protocol compliance verified through **MCP Inspector**.

EDUCATION

- Bachelor's Degree (B. E) in Electronics and Telecommunication Engineering** June 2019 – August 2023
Dr Ambedkar Institute of Technology | CGPA: 8.82 Bengaluru, India

CERTIFICATES

- Microsoft Associate AI Engineer (AI-102), Microsoft** July 2025 – August 2025
- Microsoft Azure Fundamentals (AZ-900), Microsoft** May 2024 – June 2024
- Google Professional Data Analytics Course, Coursera** June 2023 – July 2023